

Syed Muhammad Abuzar Rizvi

smabuzarrizvi@khu.ac.kr |  smabuzarrizvi

SUMMARY

Ph.D. candidate specializing in quantum computing, quantum machine learning, hybrid quantum–classical algorithms, and quantum communication, with a strong publication record in peer-reviewed journals and international conference proceedings.

SKILLS

- Programming Skills & Tools: C/C++, Python, TensorFlow, Keras, Qiskit, PennyLane, QuTiP, CPU/GPU-based quantum simulations, MATLAB, LaTeX.
- Experienced with quantum computing on various modalities, quantum machine learning, quantum circuits, variational quantum algorithms, secure quantum communication, and fundamental quantum information science theory.
- 5 years of leadership experience crafting research proposals, project management, and mentoring researchers.
- Ability to collaborate with other team members and work independently to solve complex problems.

EXPERIENCE

- **Kyung Hee University (CQILAB)**

Graduate Research Associate

Mar. 2021–Present

Yongin-si, South Korea

 - Developed machine-learning-based frameworks for quantum state classification, state discrimination, and quantum error mitigation on noisy intermediate-scale quantum devices.
 - Engineered variational quantum circuits on IBM Quantum backends for hybrid quantum–classical optimization, including VQE- and QUBO-based formulations for network scheduling and NP-hard decision problems.
 - Designed and evaluated quantum algorithms for next-generation intelligent networks, with applications in edge intelligence, resource optimization, and secure quantum communication.
 - Investigated reinforcement-learning-based quantum control, pulse-level gate design, and anonymous quantum semantic communication protocols for reliable and privacy-preserving quantum networks.
 - Led and contributed to high-impact journal manuscript preparation, including technical writing, result analysis, figure design, reviewer-response preparation, and submissions to top-tier journals.
 - Contributed to funded research programs through proposal writing, project planning, technical documentation, progress reports, and research outcome preparation.
 - Supported lab operations by coordinating research activities, mentoring junior students, assisting new student admissions and onboarding, and guiding students in experiments, implementation, and academic writing.
- **Microwave Engineering Research Lab (NUST)**

DSP Engineer

Oct. 2018–Feb. 2021

Rawalpindi, Pakistan

 - Architected hardware–software co-design solutions on Xilinx Zynq-7000 SoC platforms, programming the embedded ARM Cortex-A9 processor in C for low-latency hardware control and real-time data orchestration.
 - Engineered high-throughput RTL modules in Verilog and VHDL using Vivado, implementing real-time DSP pipelines for FIR filtering, FFT processing, fixed-point conversion, and DDR memory interfacing.
 - Designed and integrated high-speed mixed-signal processing interfaces for ADC/DAC data acquisition, SMA-based signal routing, and clock synchronization in microwave system testbeds.
 - Translated signal-processing algorithms into deployable FPGA-based hardware implementations, ensuring robust system integration, timing closure, and reliable real-time performance.

EDUCATION

- **Kyung Hee University (KHU)**

Ph.D. in Electronics and Information Convergence Engineering

Mar. 2021–Present

Yongin-si, South Korea

 - GPA: 4.09/4.30
- **National University of Sciences and Technology (NUST)**

Bachelor of Science in Electrical Engineering

Sep. 2014–Jun. 2018

Rawalpindi, Pakistan

 - GPA: 3.50/4.00

PROPOSAL WRITING & PROJECT EXPERIENCE

[Self Principle Investigator]

- **Hybrid Quantum-Classical Computation for Optimization and Machine Learning**
Funding Agency: Brain Korea 21, South Korea (2021–2022)
- **Variational Quantum Computing for Optimization: Bridging Quantum Computing, Machine Learning, and Quantum Algorithms**
Funding Agency: Brain Korea 21, South Korea (2023–2024)

[Team Member]

- **Convergence Engineering Institute for Emerging Communication Technology**
Funding Agency: National Research Foundation of Korea (2021–2023)
- **Development of 6H Next-Generation Mobile Communications Technology**
Funding Agency: Institute of Information & Communications Technology Planning & Evaluation (2021–2026)
- **Foundations for Quantum Networking with Quantum Enigmata**
Funding Agency: National Research Foundation of Korea (2021–2024)
- **Wi-Qi BRL: Toward Integrating 6G Wireless and 2G Quantum**
Funding Agency: National Research Foundation of Korea (2022–2025)
- **BK21 Education and Research Group for Innovative Human Resources in Convergence Future Communications**
Funding Agency: National Research Foundation of Korea (2022–2027)
- **Graduate School of Convergence Security, Kyung Hee University**
Funding Agency: Institute of Information & Communications Technology Planning & Evaluation (2024–2025)
- **KGCC International Joint Research Group for Computing-Converged Next-Generation Communications**
Funding Agency: Kyung Hee University (2023–2024)
- **Foundations for Computing-Intensive Quantum Communications: Toward Meta-Privacy**
Funding Agency: National Research Foundation of Korea (2025–2027)
- **Research Promotion and Competitiveness Enhancement of the Quantum Information Technology Application Research Institute, Global Software Convergence Research Institute**
Funding Agency: Kyung Hee University (2025–2026)

PUBLICATIONS AND PATENTS

Journal Papers

1. A. Khan, **S.M.A. Rizvi**, T.Q. Duong, E.K. Hong, H. Shin, “Quantum Full-Duplex Anonymous Communication in Adversarial Quantum Networks,” *IEEE Transactions on Network Science and Engineering*, 2026. (IF = 7.9)
2. H.P.Y. Duong, **S.M.A. Rizvi**, B. McNiven, T.T. Nguyen, H. Shin, O.A. Dobre, T.Q. Duong, “Quantum Machine Learning for Drug Discovery: Taxonomy, Research Challenges, and the Road Ahead,” *ACM Computing Surveys*, vol. 58, no. 8, pp. 1–36, 2026. (IF = 28.0)
3. U. Khalid, U.I. Paracha, **S.M.A. Rizvi**, H. Shin, “Quantum Computing for Intelligent Transportation Systems: VQE-Based Traffic Routing and EV Charging Scheduling,” *Mathematics*, vol. 13, no. 17, pp. 2761, 2025. (IF = 2.2)
4. **S.M.A. Rizvi**, U.I. Paracha, U. Khalid, K. Lee, H. Shin, “Quantum Machine Learning: Towards Hybrid Quantum-Classical Vision Models,” *Mathematics*, vol. 13, no. 16, pp. 2645, 2025. (IF = 2.2)
5. **S.M.A. Rizvi**, U. Khalid, S. Chatzinotas, T.Q. Duong, H. Shin, “Controlled Quantum Semantic Communication for Industrial CPS Networks,” *IEEE Transactions on Network Science and Engineering*, 2025. (IF = 7.9)
6. J.U. Rehman, **S.M.A. Rizvi**, S. Koudia, S. Chatzinotas, H. Shin, “MIMO Quantum Communication in Correlated Pure-Loss Channels,” *IEEE Communications Letters*, vol. 29, no. 7, pp. 1544–1548, 2025. (IF = 4.4)
7. **S.M.A. Rizvi**, N. Asif, M.S. Uloom, T.Q. Duong, H. Shin, “Multiclass Classification of Metrologically Resourceful Tripartite Quantum States with Deep Neural Networks,” *Sensors*, vol. 22, no. 18, pp. 6767, 2022. (IF = 3.5)

Conference Proceedings

1. **S.M.A. Rizvi**, J.U. Roh, H. Shin, “Deep Reinforcement Learning for Optimal Pulse Shaping in Single-Photon Quantum Emitters,” in *40th International Conference on Information Networking (ICOIN)*, pp. 180–184, 2026.
2. E. Gholamalishahiharouni, A.W. Shaffar, **S.M.A. Rizvi**, U. Khalid, H. Shin, “Analytical Design of an LC Double-Notch Purcell Filter for Superconducting Quantum Circuits,” in *Korea Information and Communications Society Conference Proceedings*, pp. 330–331, 2025 **Best Paper Award**.
3. M.M.U. Gondel, **S.M.A. Rizvi**, U. Khalid, H. Shin, “NISQ State Preparation with Reinforcement Learning,” in *Korea Information and Communications Society Conference Proceedings*, pp. 950–951, 2025.
4. U.I. Paracha, **S.M.A. Rizvi**, M.M.U. Gondel, W. Park, H. Shin, “Adaptive Quantum Readout Error Mitigation with Transfer Learning,” in *15th International Conference on Information and Communication Technology Convergence*, 2024.

5. **S.M.A. Rizvi**, M.M.U. Gondel, U.I. Paracha, H. Shin, "Variational Quantum Eigensolver for Optimizing Network Scheduling Using QUBO Formulation," in *International Conference on Industrial Networks and Intelligent Systems*, pp. 46–57, 2024.
6. **S.M.A. Rizvi**, M.S. Ulum, N. Asif, H. Shin, "Neural Networks with Variational Quantum Circuits," in *International Conference on Industrial Networks and Intelligent Systems*, pp. 203–214, 2023 **Best Paper Award**.
7. **S.M.A. Rizvi**, F. Zaman, K. Lee, H. Shin, "Quantum Optimal Control with Pulses," in *Korea Information and Communications Society Conference Proceedings*, pp. 153–154, 2022.
8. **S.M.A. Rizvi**, M.A. Ullah, H. Shin, "Measurement Error Mitigation for NISQ Devices," in *Korea Information and Communications Society Conference Proceedings*, pp. 1393–1394, 2022.
9. F. Zaman, **S.M.A. Rizvi**, H. Shin, "Quantum-Assisted Reinforcement Learning," in *Korea Information and Communications Society Conference Proceedings*, pp. 516–518, 2022.
10. **S.M.A. Rizvi**, M.A. Ullah, H. Shin, "Machine Learning for Qubit State Estimation on IBM Quantum Computer," in *Korea Information and Communications Society Conference Proceedings*, pp. 472–474, 2022. **Best Paper Award**

Patents

1. H. Shin, **S.M.A. Rizvi**, F. Dzulfikar, M.S. Ulum, J.U. Roh, "Apparatus and Method of Mitigating Errors in Quantum Fourier Transform," *Korea Patent Application No. 10-2023-0158507* [submitted].
2. H. Shin, H.J. Jung, M.S. Choi, F. Zaman, **S.M.A. Rizvi**, S.A. Wachid, F. Dzulfikar, "Method of Error Mitigation for Noisy Intermediate-Scale Quantum (NISQ) Computation," *Korea Patent Application No. 10-2023-0002208* [submitted].
3. H. Shin, J.U. Roh, U. Khalid, **S.M.A. Rizvi**, T. Muhammad, "Quantum Reservoir Learning for Reinforcement-Learning-Optimized Pulse and Hamiltonian Control System and Method for a Rydberg Lattice," *Korea Patent Application No. 10-2026-0044766* [submitted].
4. H. Shin, **S.M.A. Rizvi**, A. Naema, U. Khalid, M.A. Ullah, A. Khan, "Apparatus and Method for Classifying Quantum States in Quantum Information Integrated Processing System," *Korea Patent No. 10-2820473*.

CERTIFICATIONS

- **Qiskit Global Summer School on Quantum Machine Learning**

July 2021

REFERENCES

Available upon request.